

THE TWO-WAY LINE 京张对开

Running the Centennial Jing-Zhang Corridor in Both Directions — an AI-belt urban design



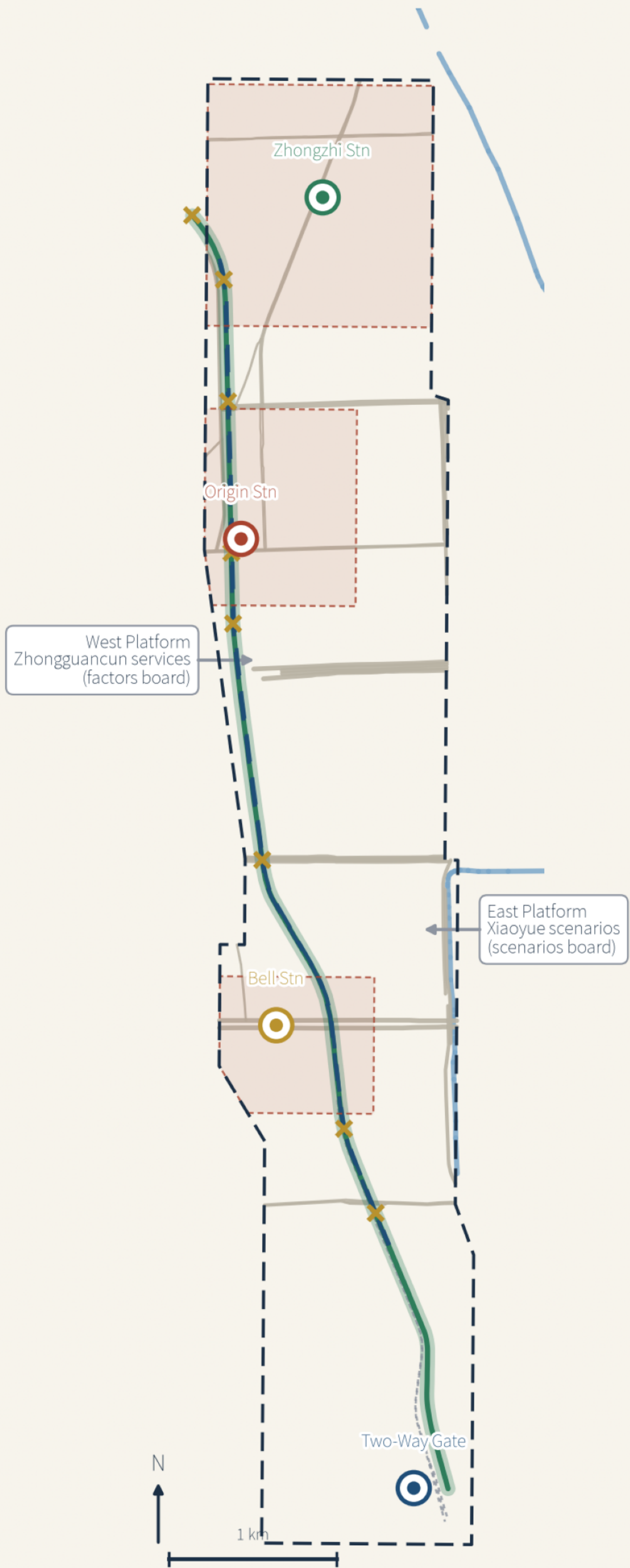
Identity • Two arrows, one track

Two parallel rails bound by sleeper bars, north-pointing above and south-pointing below; the negative space between them forms the skeleton of the character 开 ("open"). Green = up-line energy and compute, deep blue = the Under Line, brick red = heritage and sleepers.

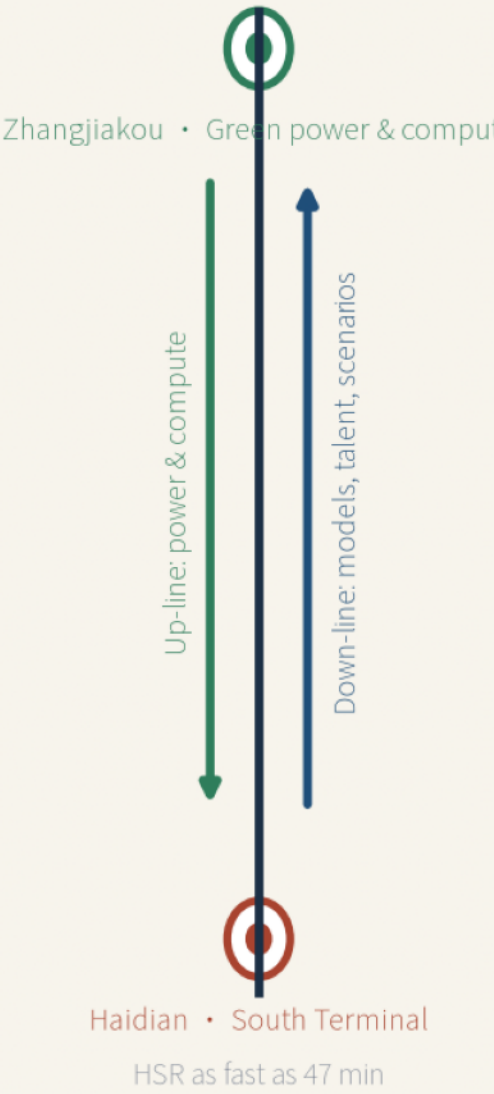
Master Concept Board

THE TWO-WAY LINE • Master Concept

Surface + Under Lines • 3 Stations 1 Gate • 2 Platforms 8 Crossings • Twin-River Cooling Loop — 11.4 km²



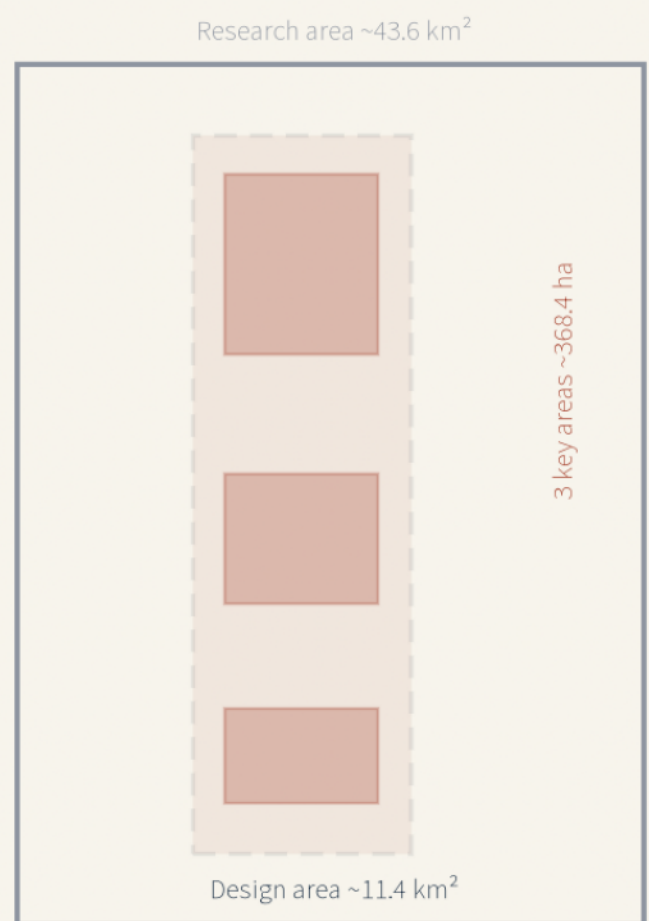
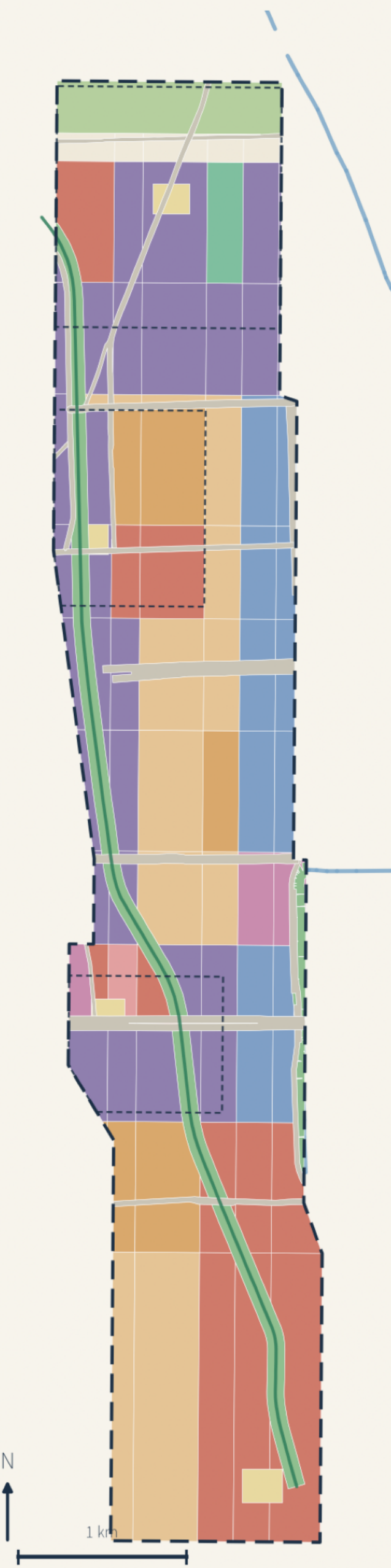
The Two-Way Corridor (concept)



- Surface Line • 7.8 km heritage slow spine
- Under Line • Qinghuayuan HSR tunnel (indicative)
- Design area (provisional)
- Key areas (provisional)
- Crossings (8 east-west stitches)
- Qinghe / Xiaoyue R.
- Metro Line 13

Three-Level Scope & Land Use

43.6 km² research • 11.4 km² design • 368.4 ha key areas — 189-parcel full partition



Research	258.0 × 10³ m² 22.0%
Housing	209.3 × 10³ m² 18.4%
Commercial	203.5 × 10³ m² 17.9%
Education	111.5 × 10³ m² 9.8%
Community svc.	89.9 × 10³ m² 7.9%
Park green	80.3 × 10³ m² 7.0%
Roads	74.1 × 10³ m² 6.5%
Protective green	40.6 × 10³ m² 3.6%
Culture	20.8 × 10³ m² 1.8%
White land (reserve)	18.6 × 10³ m² 1.6%
Sports	15.4 × 10³ m² 1.4%
Plaza	11.7 × 10³ m² 1.0%
Health	6.3 × 10³ m² 0.6%

Core thesis

Read 'Jing-Zhang' in full — the corridor has two ends. In 1909 the railway carried energy and engineering confidence into Beijing; in 2019 the HSR kept racing beneath the heritage park through Qinghuayuan Tunnel; in 2022 'Eastern Data, Western Computing' placed training compute on Zhangjiakou's green power. The belt is not a 9-km monologue but the southern terminal of a 200-km two-way relationship: power and compute up-line, models, talent, and scenarios down-line.

Structure • Surface+Under Lines / 3 Stations 1 Gate / 2 Platforms 8 Crossings / Cooling Loop

Surface Line = the 7.8-km heritage slow spine for people and culture; Under Line = the HSR in the tunnel for speed and reach. Three stations = Zhongzhi (training & standards), Origin (campus-city switch-back), Bell (content & data assets); one gate = the Two-Way Gate at Beijing North. West Platform = Zhongguancun services; East Platform = Xiaoyue scenarios; eight crossings stitch east and west. The Twin-River Cooling Loop joins the Qinghe and Xiaoyue with cabin waste-heat reuse.

Institutional core • the Two-Way Protocol

Every AI deployment is a paired run: an up-line run that collects data and consumes compute must be matched by a down-line run delivering services and public returns; who can stop it and how to exit is written on every card. Extraction-only deployments are never scheduled. Releases follow an annual public Timetable.

Three key areas • the density gradient from north to south

Key area	Area	Buildings	Concept GFA	Concept gross PR	Leading land use	Character
Zhongzhi • Zhongzhiyuan	192.1 ha	47	1.448 M m²	0.75	research 43.1%	campus
Origin • AI Origin Community	104.3 ha	49	1.005 M m²	0.96	community 35.7%	district
Bell • Dazhongsi	72.0 ha	38	1.208 M m²	1.68	research 59.3%	urban

Concept gross plot ratio is concept floor area divided by key-area area — not a statutory plot ratio. Land use is recomputed parcel-by-parcel against each key area in EPSG:4548 and covers each area exactly.

Two scope facts • disclosed by us

(1) Of the eight crossings, Qinghe Bay sits about 85 m beyond the northern edge of the overall design area — its job is to carry the Surface Line past the boundary to the Qinghe waterfront, so crossing is functionally necessary and its land is excluded from in-area recomputation. (2) Qinghuayuan Station is about 389 m west of the area, inside the coordinated research scope, so run S8 is a deliberately cross-scope ceremonial venue. Both were found by our own recomputation and registered in assumptions.json.

11.40 km² design area (provisional, +0.0006%)

368.4 ha three key areas (±0.01%)

7.8 km through slow spine

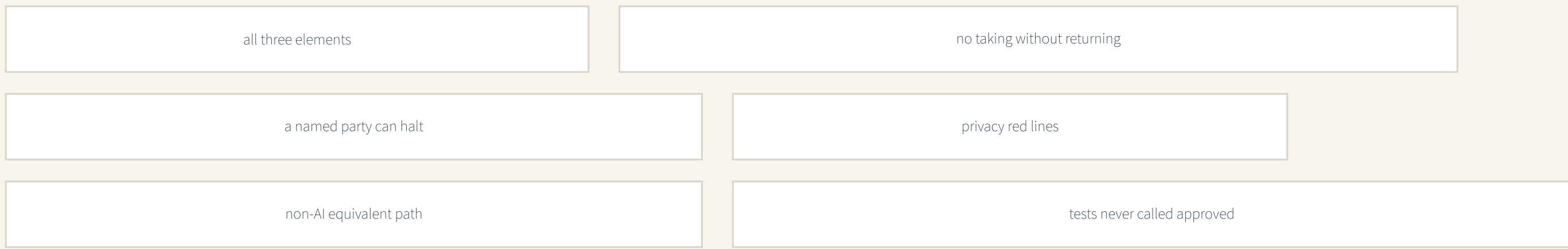
8+3+1 12 scenario nodes

12/3 cards / industry tests

14 renewal projects, 3 phases

Two-Way Protocol • a recomputable admission contract

Six rules taken line by line from this proposal, encoded as a machine-readable contract and a tabletop exercise; reviewers can run it themselves rather than take the text on trust.



12 / 12 schedulable

Twelve service cards

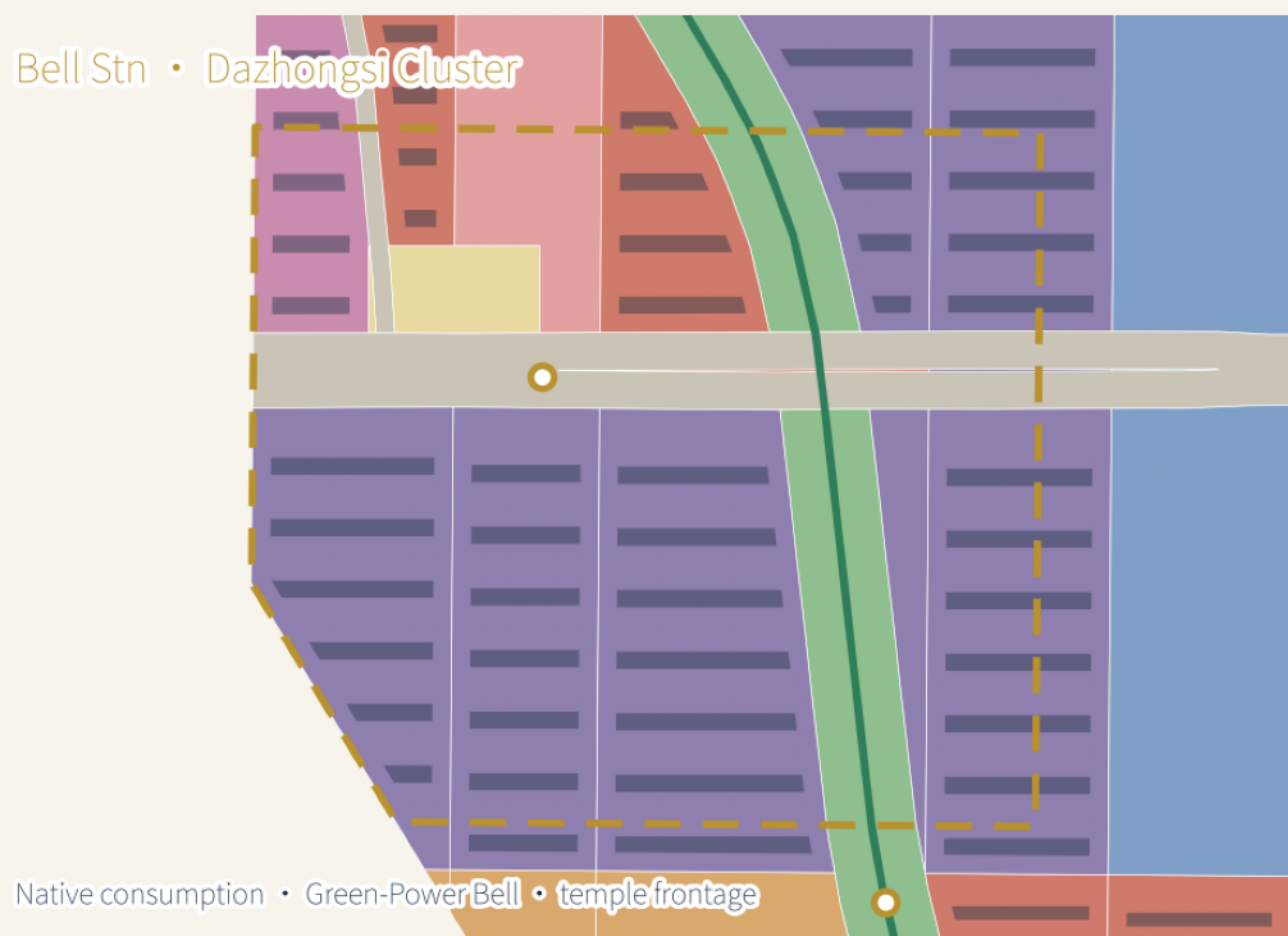
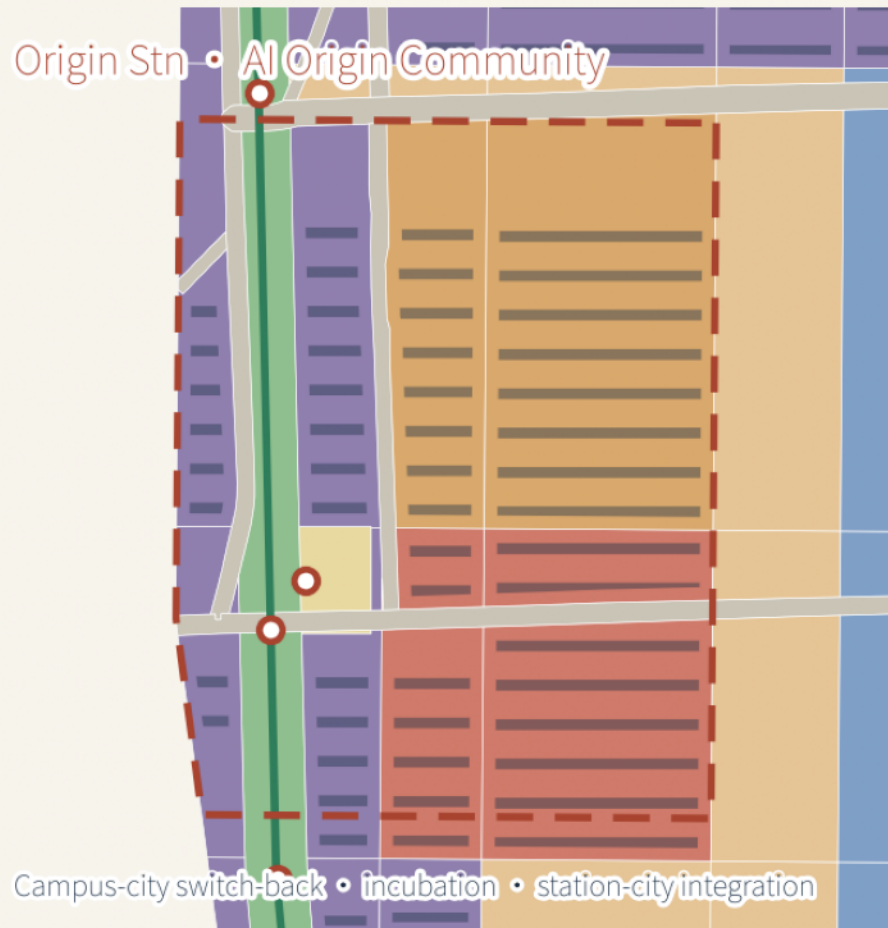
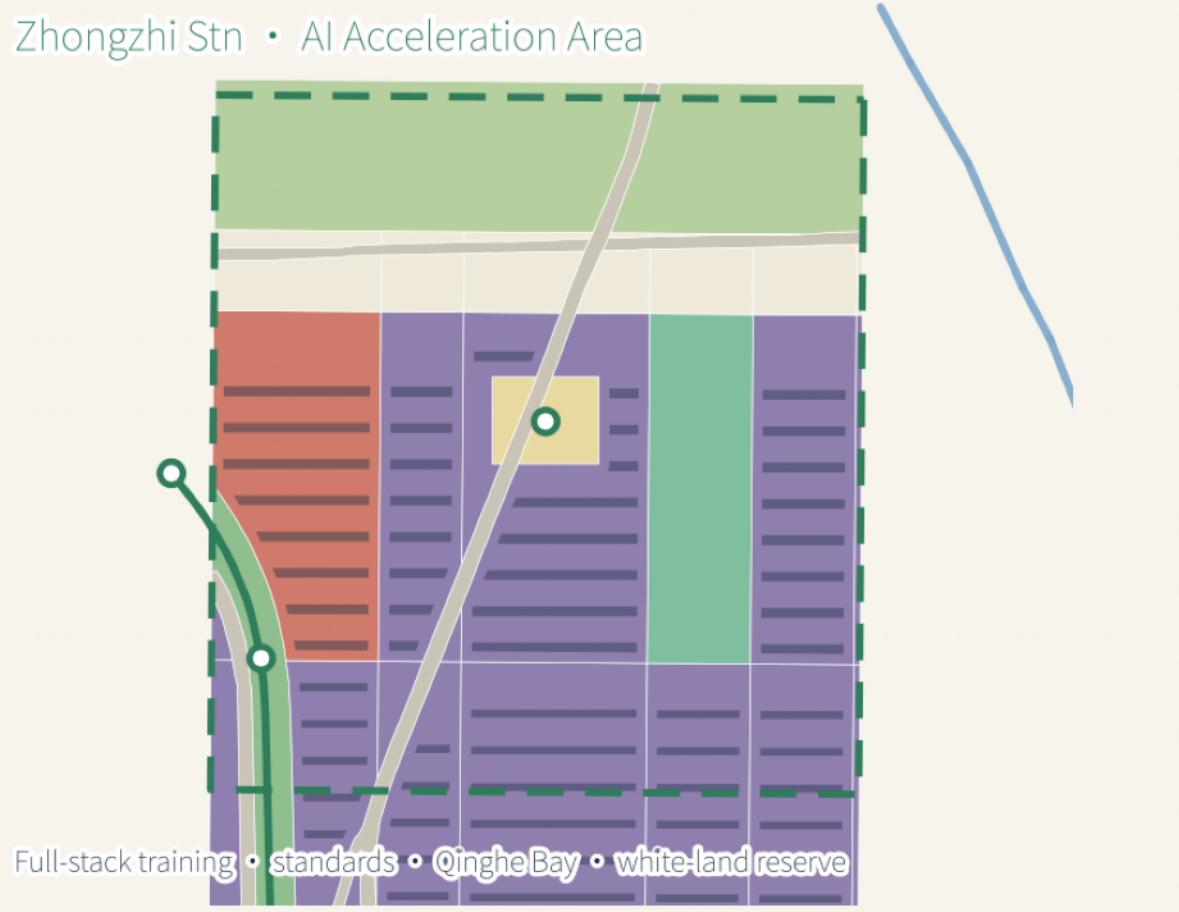
6 / 6 caught by their rule

Deliberately broken runs

node visual/assets/run_twoway_tabletop.js • Node 18+, zero dependencies

Detailed Design of Three Key Areas

Zhongzhi (192.1 ha) • Origin (104.3 ha) • Bell (72.0 ha) — implementation-scheme depth (concept)



Boundary status: provisional rough frame, self-derived from OSM centerlines of announced streets and area-calibrated, not an official redline; no approval/ownership precise area use full recalculation when official polygons arrive.
Base data © OpenStreetMap contributors (ODBL). Derived from packaged GeoJSON + metrics, concept proposal, w943 / jingzhang two-way line • 2020-08

Mobility & Blue-Green Systems

14 micro-circulation streets • through spine • 8 crossings • cooling loop • 4 station hubs

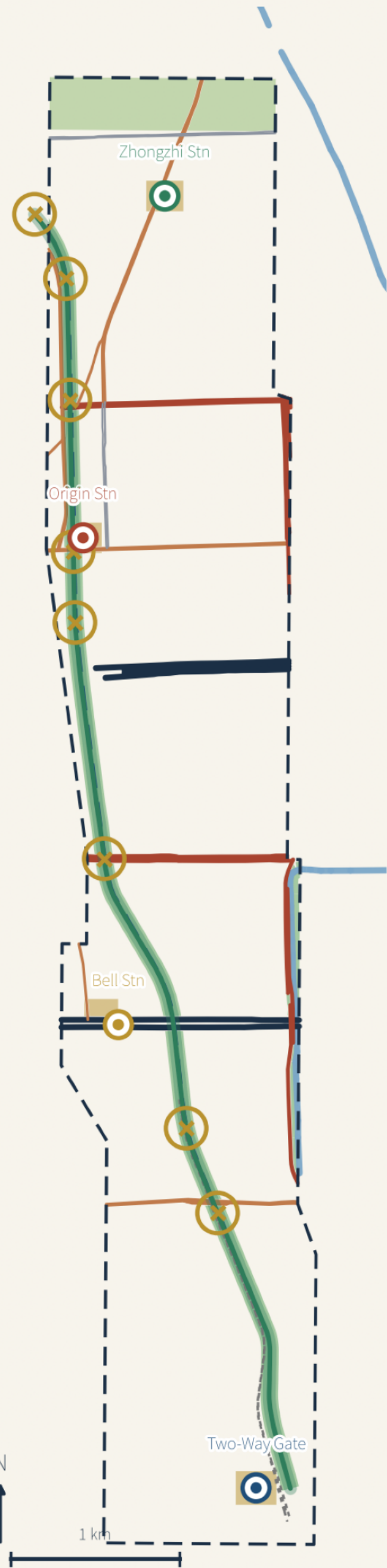
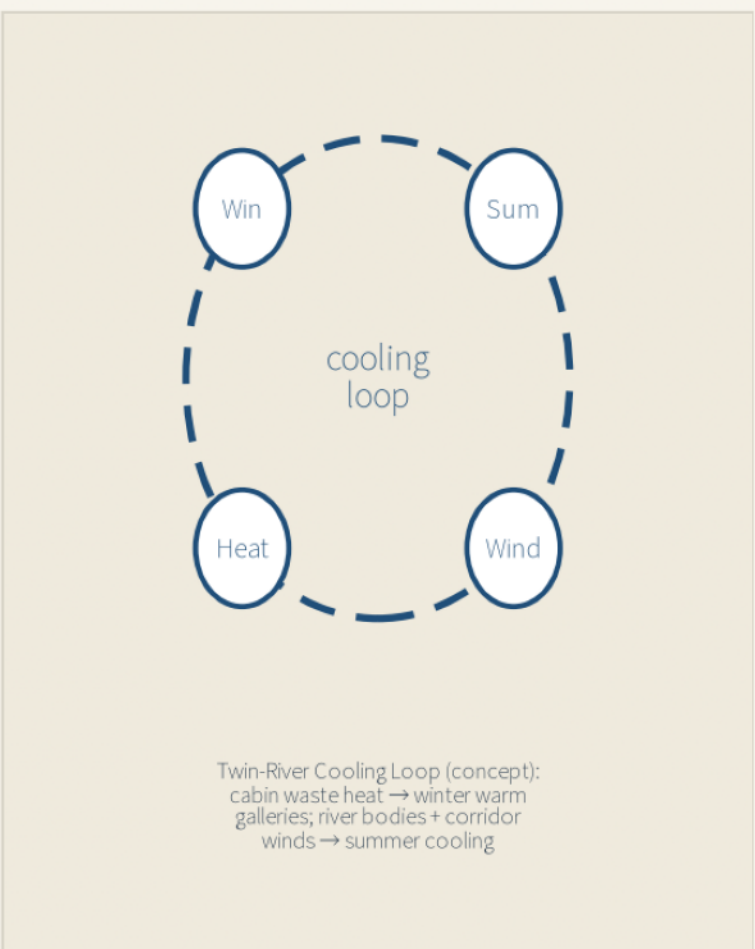
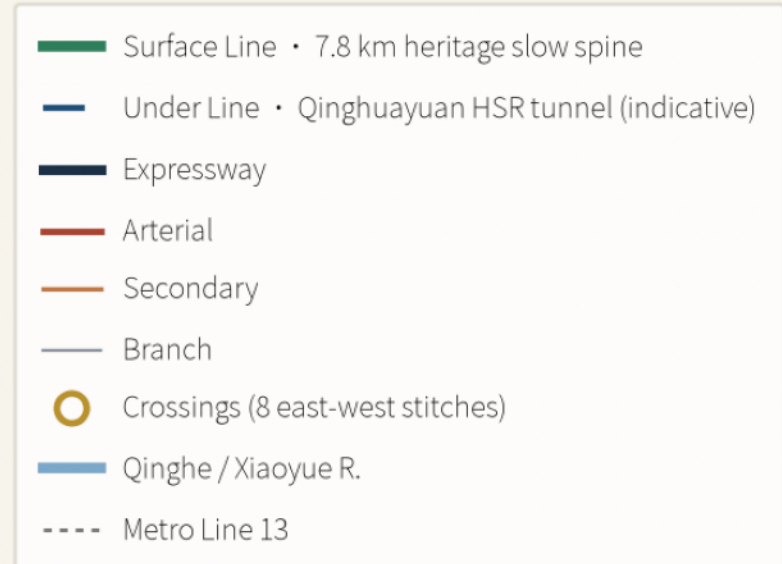


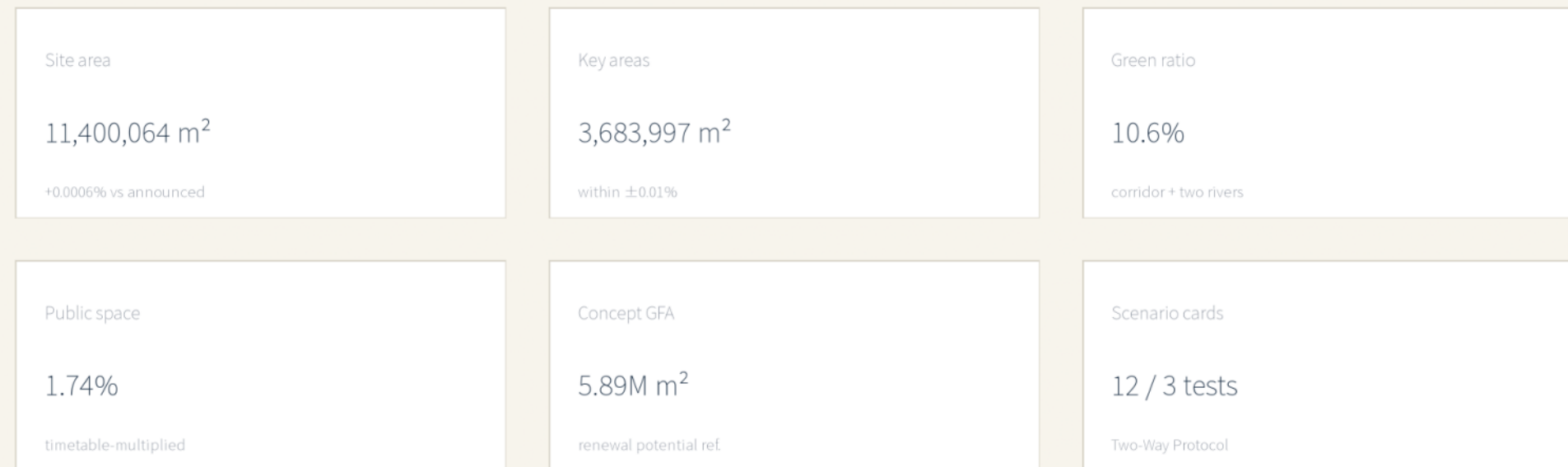
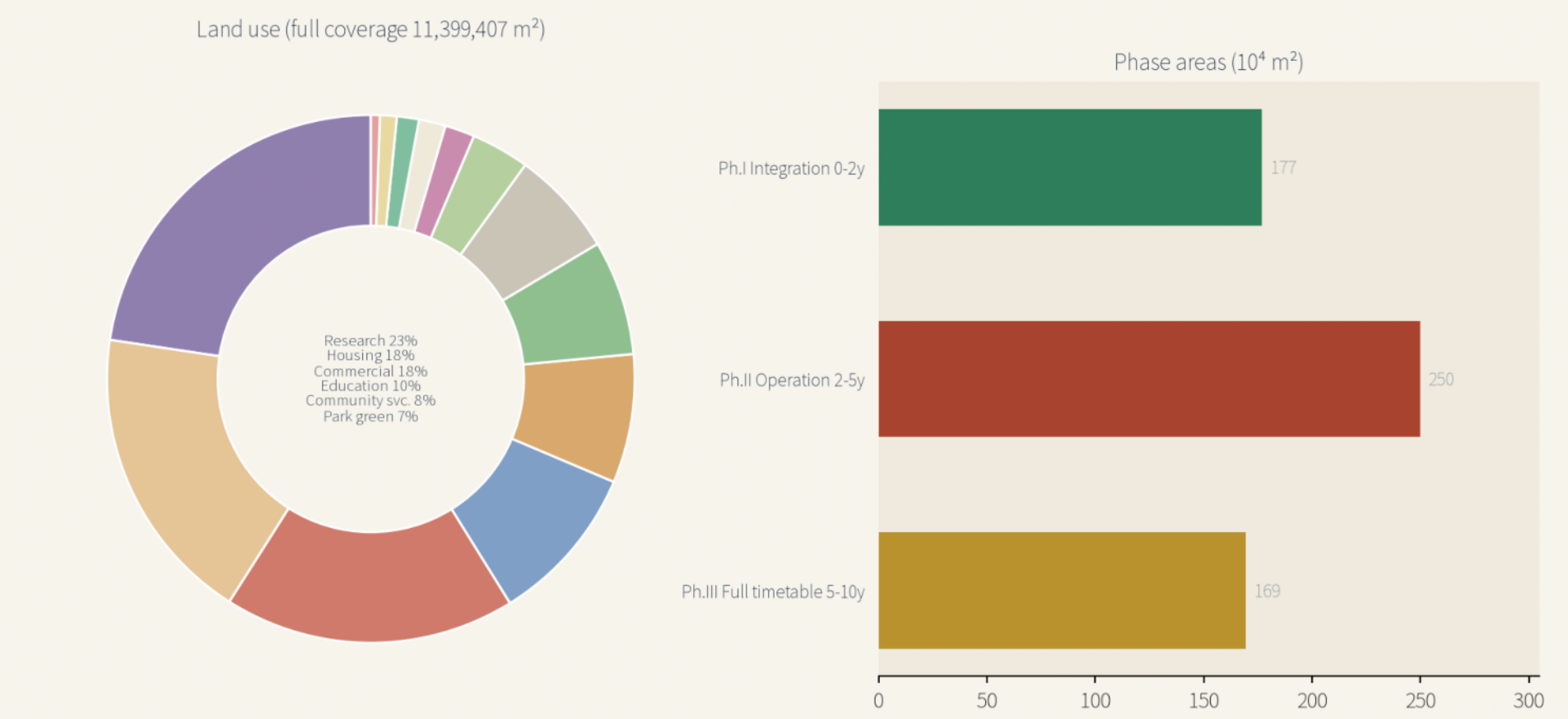
FIG-04



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Metrics Recalculation & Evidence Chain

All spatial metrics recomputed from packaged GeoJSON in EPSG:4548 • spatial review PASS



Evidence chain: every claim traces to layers and metrics



Pending official data (registered, never faked): FAR • max height • AI index baseline • talent density baseline

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Twelve run-cards (Two-Way Protocol: what goes up • what comes down • who can stop it)

Run	Scenario	Node	Up-line	Down-line	Review & exit
T1	Low-speed delivery test	Qinghe Bay	de-identified ops	free delivery hours	on-site stop
T2	Traffic dispatch test	crossings	anonymous flows	published waits	manual fallback
T3	Copilot sandbox	Bell Stn	authorized tickets	SME quota	human switchable
S4	Cultural guide	mileposts	anonymous heat	4-language guiding	human review
S5	Health navigation	Xueyuan Rd	voluntary asks	elder escort	physician review
S6	Safety ops review	the Gate	incident data	published drills	post-hoc only
S7	Waste-heat dispatch	cooling loop	telemetry	warm gallery	manual bypass
S8	Model release day	release hall	registration	open weights	community review
S9	Accessible walking	8 crossings	facility status	blind guidance	hotline backstop
S10	Campus matching	Origin hall	licensed list	shared returns	manager review
S11	Micro-climate	whole spine	sensors	heat warnings	stoppage points
S12	Green power chime	Bell plaza	power share	hourly chime	auditable source